SALES DATA ANALYSIS

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1. Introduction

This report presents a detailed analysis of a retail sales dataset with the objective of understanding customer purchasing behaviour. The analysis focuses on identifying which customer segments contribute the most to overall sales based on demographic, geographic, and professional attributes. Insights from this study can support data-driven decisions in marketing, sales strategy, and customer targeting.

2. Project Objectives

- Analyse customer spending patterns across gender, age group, state, occupation, product category, and marital status
- Identify high-value customer segments
- Generate actionable business insights using data analysis and visualization techniques

3. Dataset Overview

The dataset consists of customer-level retail transaction records stored in CSV format. Each record represents a purchase made by a customer and includes demographic details, location, occupation, product category, number of orders, and sales amount.

Key Columns:

- Gender, Age, Age Group, Marital Status
- State, Zone
- Occupation
- Product Category
- Orders, Amount

The primary metric used for analysis is **Sales Amount**, which represents the total value of purchases.

4. Tools & Technologies Used

- Pandas and NumPy for data cleaning and analysis
- Matplotlib and Seaborn for data visualization
- Google Colab for execution and reporting

5. Data Cleaning & Preparation

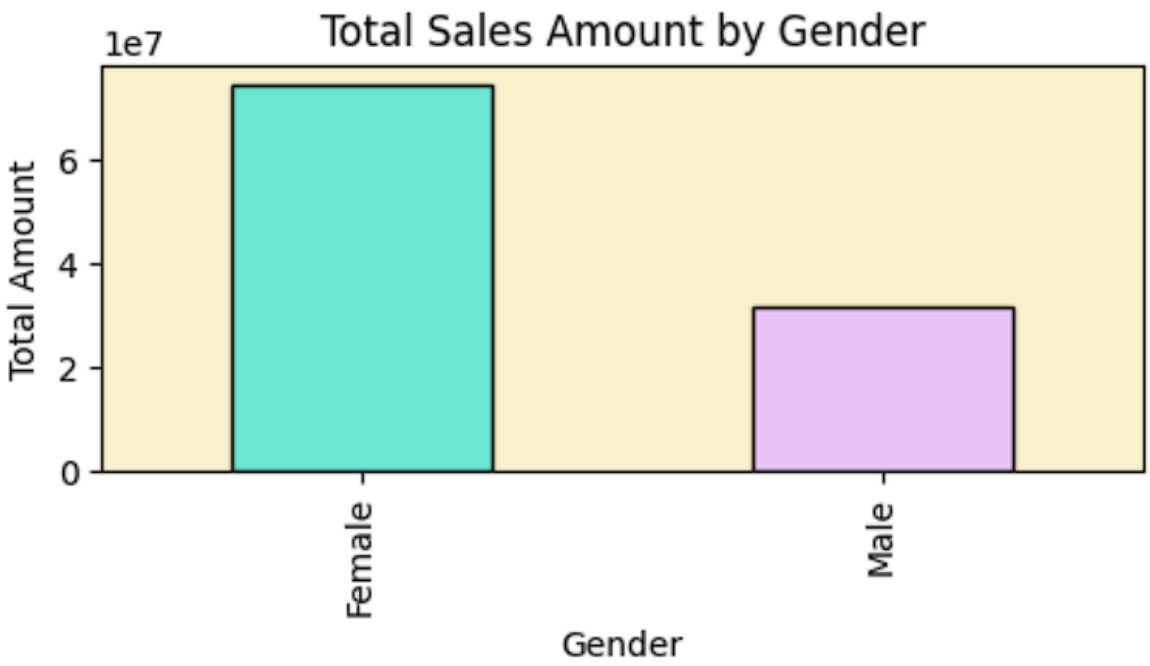
The following steps were performed to prepare the data for analysis:

- Removed unnecessary columns such as Status and unnamed1
- Checked and handled missing values in the Amount column
- Renamed columns for better readability (e.g., Marital_Status to Marital)
- Standardized categorical values:
    - Gender: F → Female, M → Male
    - Marital Status: 0 → Single, 1 → Married
- Ensured consistency and accuracy across all columns

6. Exploratory Data Analysis (EDA)

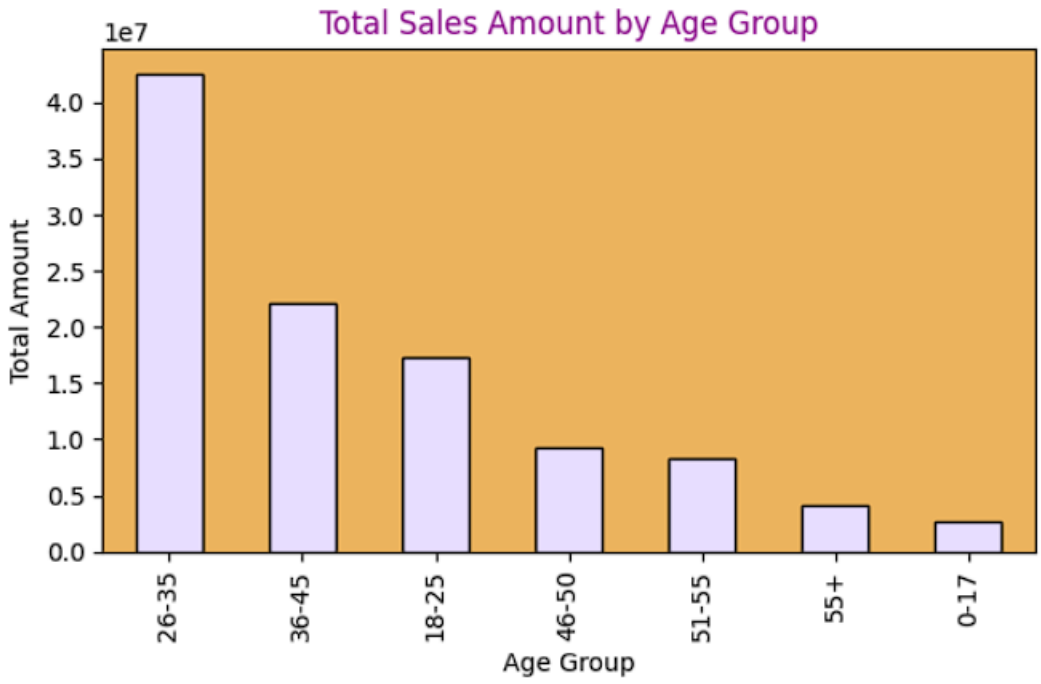
6.1 Gender-wise Sales Analysis

A comparison of total sales by gender was conducted using bar charts. The analysis shows that **female customers contribute a higher share of total sales compared to male customers**.



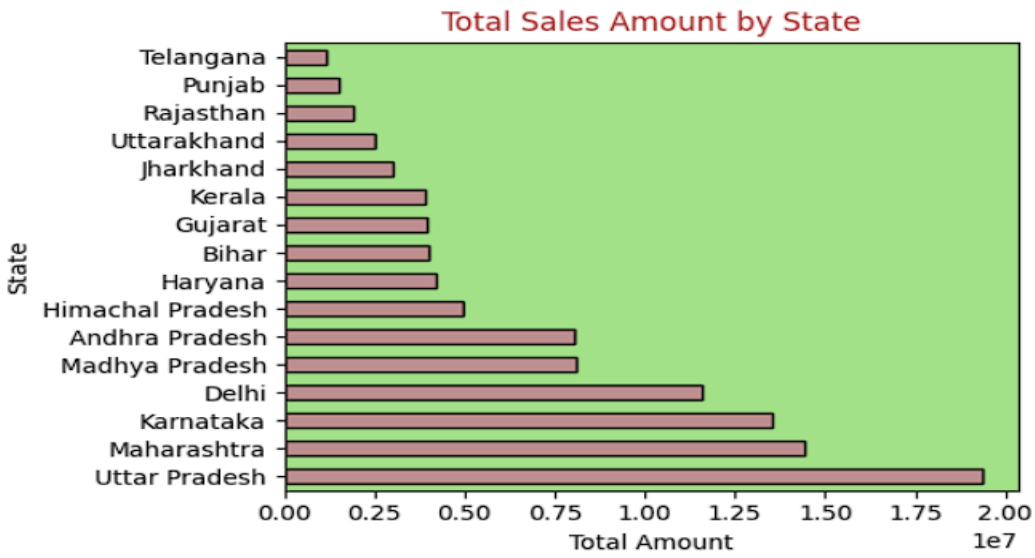
6.2 Age Group-wise Sales Analysis

Sales performance was analysed across different age groups. Customers in the **26–35 age group** were identified as the highest spenders, indicating strong purchasing power among young working professionals.



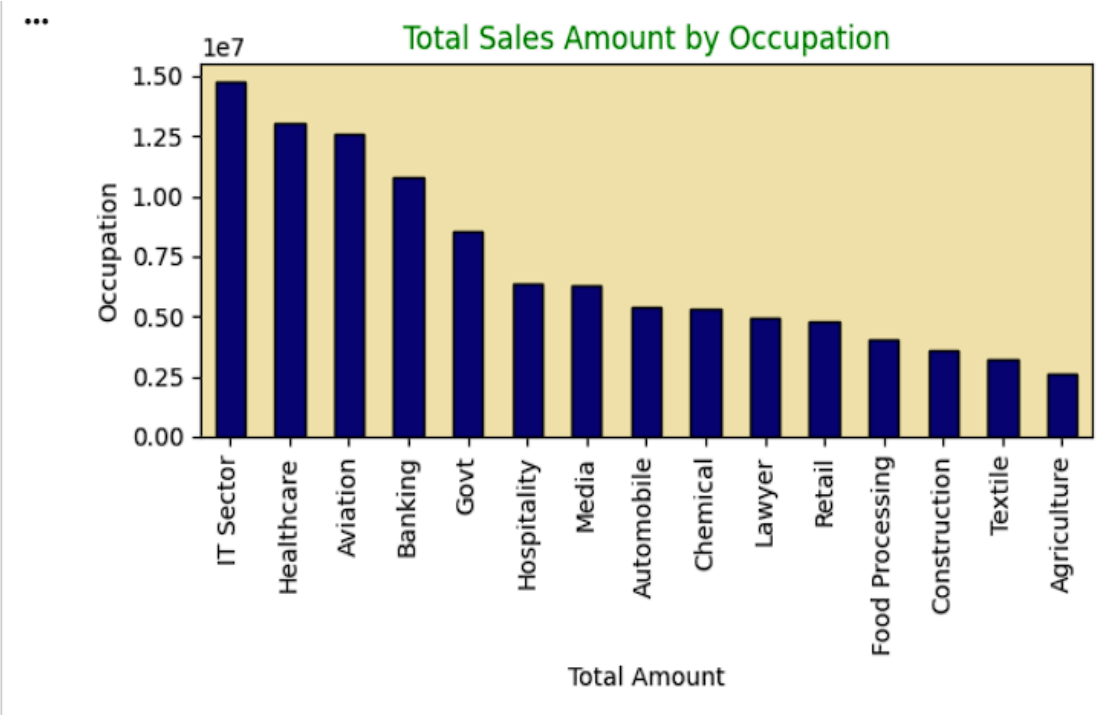
6.3 State-wise Sales Performance

State-level analysis revealed that **Uttar Pradesh, Maharashtra, and Karnataka** generate the highest sales, making them key regions for business growth and targeted marketing.



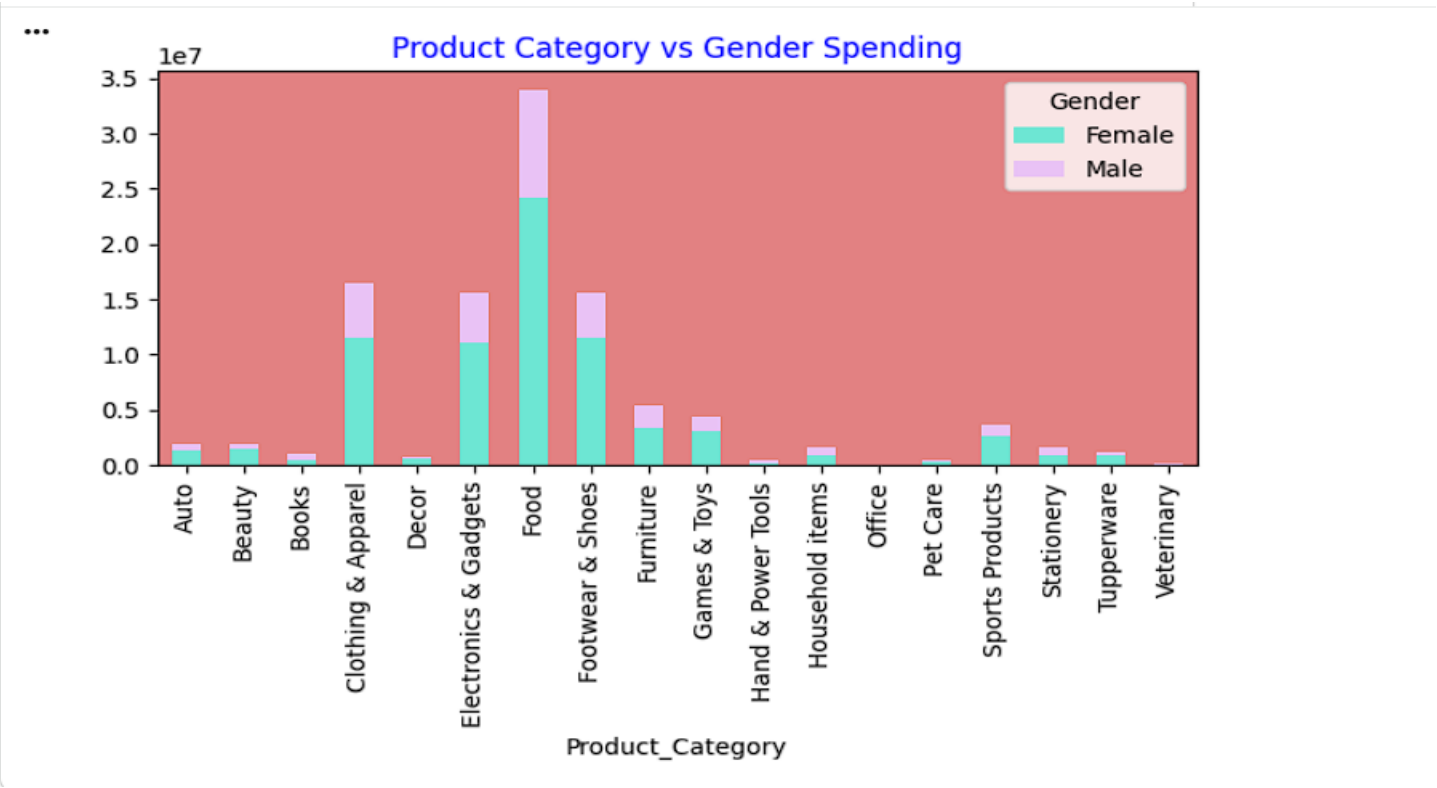
6.4 Occupation-wise Sales Analysis

Customers working in **IT, Healthcare, and Banking sectors** showed higher spending patterns, highlighting the importance of income stability in purchasing behaviour.



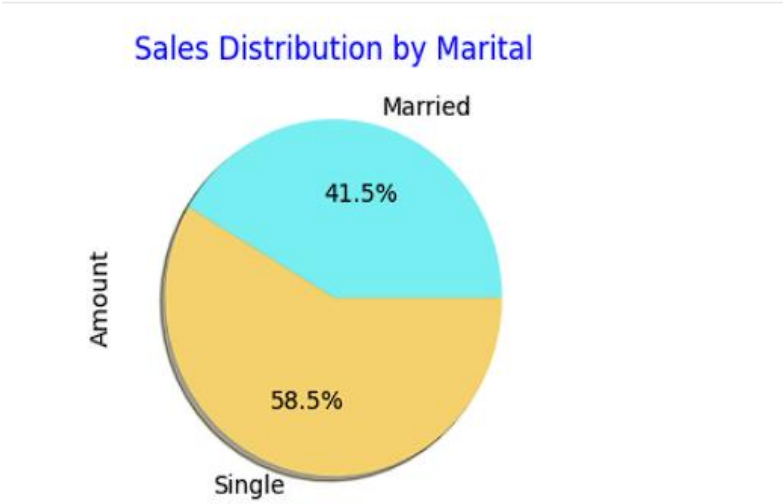
6.5 Product Category-wise Sales Analysis

Revenue contribution was analysed across product categories to identify top-performing segments. Certain categories consistently generated higher sales, indicating strong customer demand.



6.6 Marital Status-wise Sales Analysis

Spending behaviour between single and married customers was compared. The analysis indicates that **single customers spend more on average than married customers**.



7. Data Visualization & Insights

Bar charts and comparative visualizations were used throughout the analysis to clearly communicate trends and differences among customer groups. Visual analysis helped identify patterns quickly and supported meaningful interpretation of results.

8. Key Findings

- Female customers are the primary contributors to overall sales.
- The 26–35 age group represents the most valuable customer segment.
- A few key states dominate total sales performance.
- Professionals in IT, Healthcare, and Banking sectors spend more.

- Single customers demonstrate higher purchasing behaviour.

9. Business Recommendations

- Focus marketing campaigns on female customers aged 26–35.
- Launch state-specific promotions in top-performing regions.
- Design targeted offers for professionals in high-spending occupations.
- Optimize inventory based on high-performing product categories.

10. Conclusion

This Sales Data Analysis project demonstrates how structured data can be transformed into actionable business insights through effective data cleaning, exploratory analysis, and visualization. The findings provide valuable inputs for strategic decision-making in marketing, sales optimization, and customer segmentation.